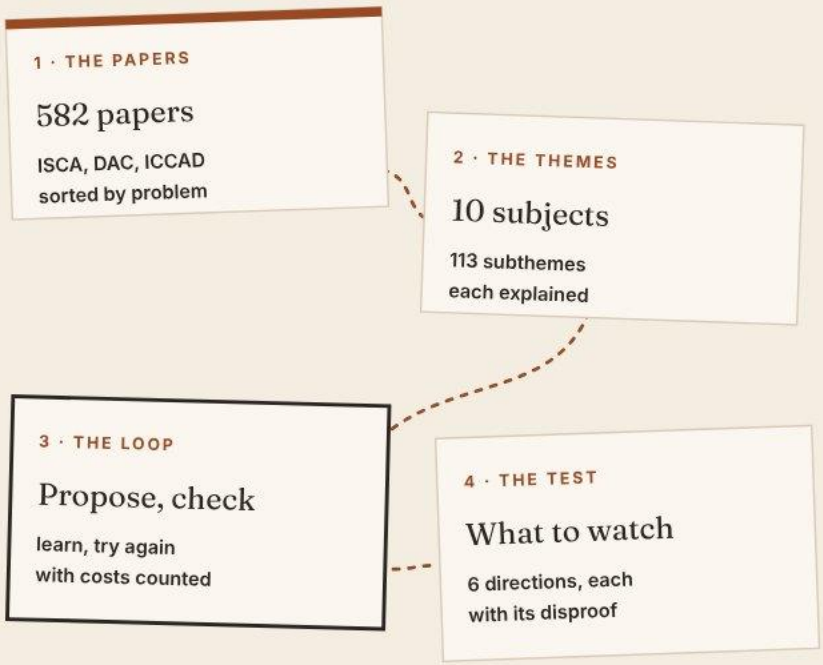
ISCA 2026 · DAC 2026 · ICCAD 2025 · 582 papers

What do this year's chip-design conferences say about where the field is going?

A plain-language atlas of the research, with what each paper shows and leaves open.



THE WHOLE STORY

In three sentences.

1 · THE DIRECTION

Chip design is becoming a loop: propose, check, learn, try again.

Six directions recur across the 582 papers. Most are about feeding checks, measurements and physical limits back into the next design choice.

2 · THE WEIGHT

Accelerators, memory and reliability draw the most papers.

Largest of 10 themes: architecture and accelerators (103), memory, interconnect and storage (85), power, reliability and security (78).

3 · THE CAUTION

Most papers were read from short descriptions, so these are hypotheses.

541 of 582 records come from program descriptions; 33 from a full paper or preprint. DAC supplies 544 of the 582.

HOW THE ATLAS IS BUILT

From papers to directions, in five steps.

582 papers, sorted into 10 themes and 113 subthemes, a few read in full, then six directions to watch.



WHERE THE PAPERS COME FROM

DAC is 544 of 582 papers.



PAPERS PER CONFERENCE

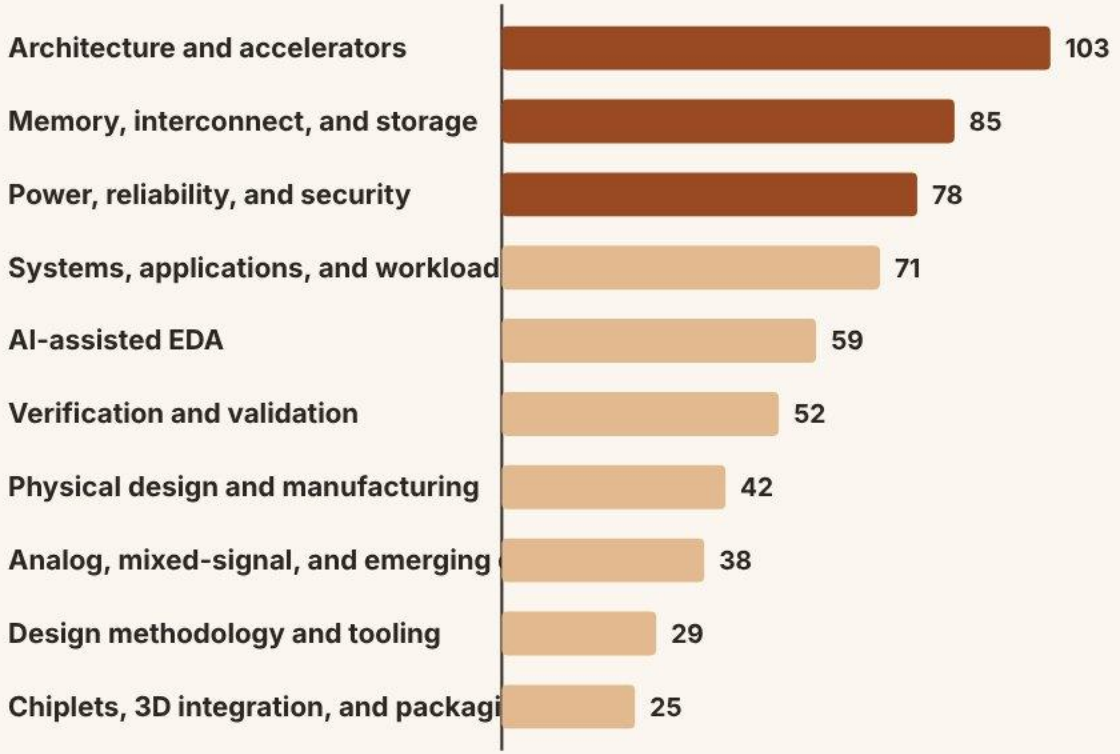
The sample leans toward design automation.

ISCA adds 17; ICCAD adds 21, from its 2025 edition.

ICCAD 2026 papers were not yet public.

WHERE THE ATTENTION GOES

Accelerators, memory and reliability lead.



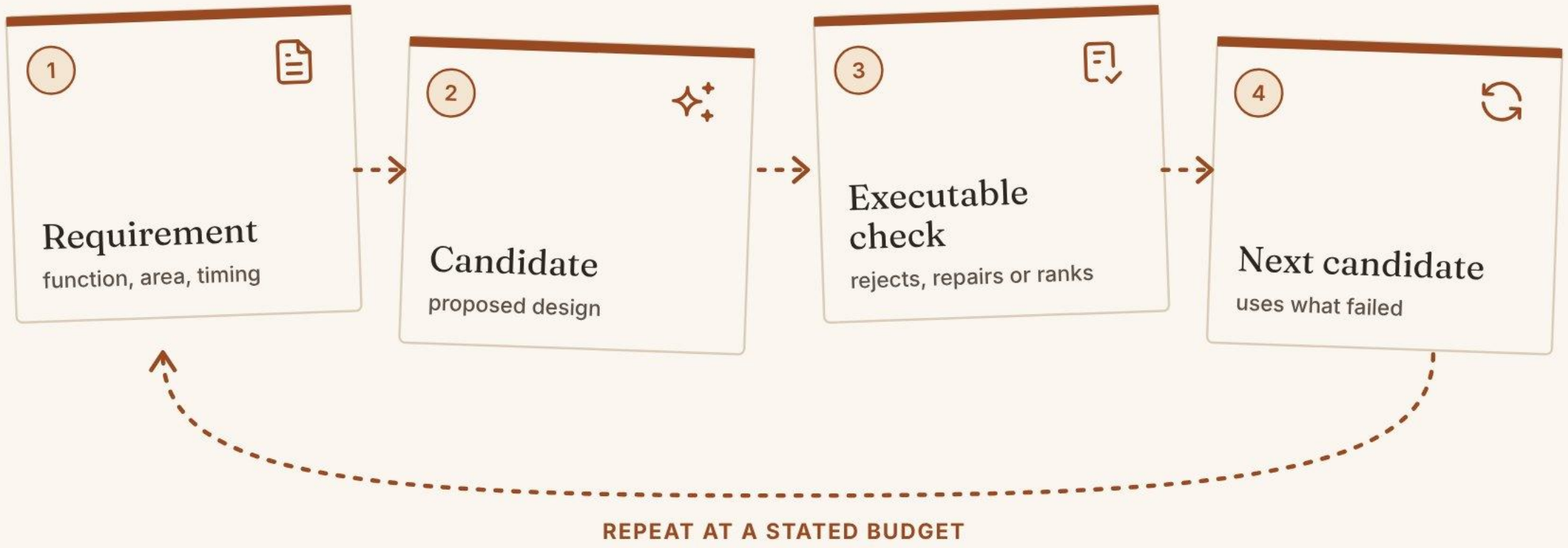
PAPERS PER THEME

Architecture and accelerators: 103 papers.

A count shows attention in this sample, not importance.

Each paper sits in one theme.

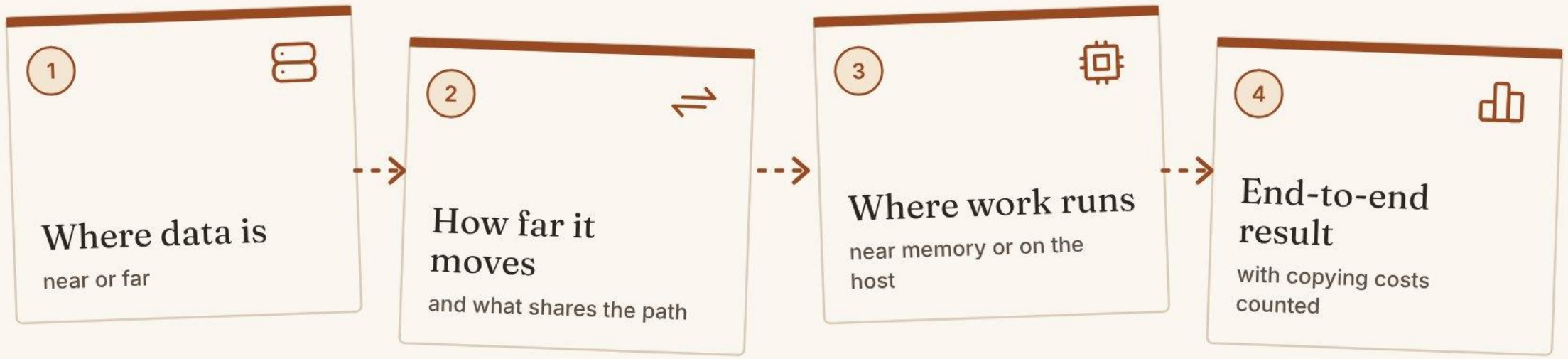
Design becomes a loop.



Checks steer the next attempt, not just judge the last one.

Moving data is often the real cost.

More arithmetic units help only when the data arrives in time.



Where data sits decides how fast and how cheaply work gets done.

Heat, wires, faults and attacks move earlier.

PHYSICAL REALITY

Heat, wire delay, power

Fast physical estimates guide early choices; an accurate check must still confirm the final design.

FAULTS AND ATTACKS

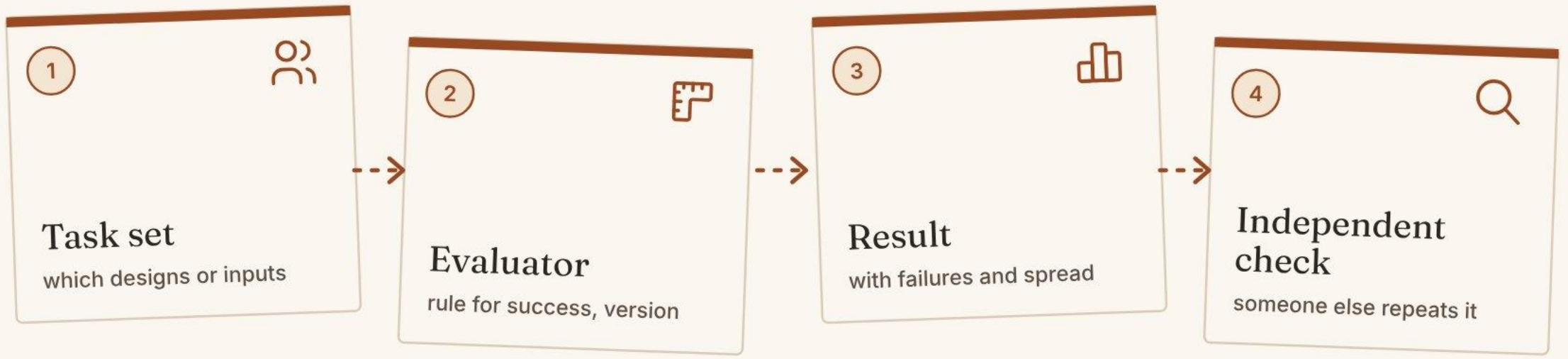
Named before protection

Protection costs speed and energy. Report useful correct work with the protection switched on.

MEASUREMENT

Better ways to measure results.

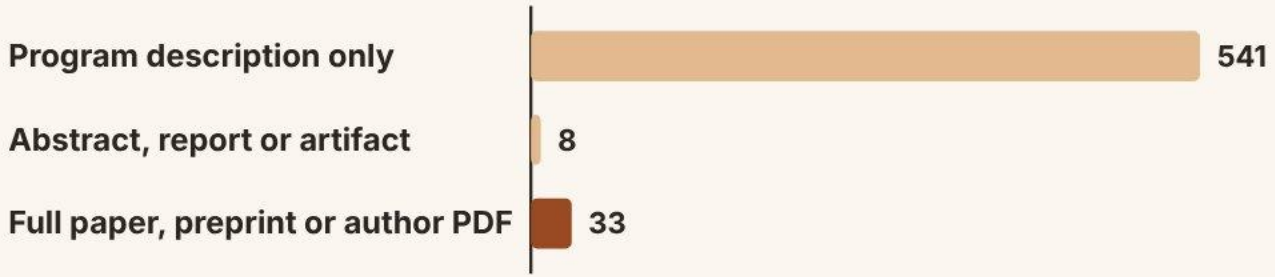
A score depends on the task set, the rule for success and the comparison.



How results are measured is becoming research in its own right.

HOW SURE WE ARE

541 of 582 records come from short descriptions.



WHAT EACH RECORD WAS CODED FROM

Most records rest on the official program description.

16 papers were checked in depth against their full source.

So the directions are hypotheses, not measured trends.

WHAT TO WATCH

Six directions, each with its disproof.

DIRECTION 1

Requirements checked while designing

DIRECTION 2

Moving data is the real cost

DIRECTION 3

Measure, then decide the next design

DIRECTION 4

Plan for faults and attacks early

DIRECTION 5

Heat, wires and power in the loop

DIRECTION 6

Better ways to measure results

YOUR TURN

Start with a theme, then follow its papers.

Every page explains the problem before the papers, and says how deep its sources go.

